We'll see in the section 4.10 how to choose the the number of hash functions and the size of the array in order to optimize the ratio of false positives for a Bloom filter capable of storing a given number of keys.

4.8.1 Why There Are No False Negatives...

In the simplest version of Bloom filters deletion is not allowed. This means that if a bit is flipped when a key is stored, then it will never ever be set to `0` again.

At the same time, the output of the hash functions is deterministic and constant in time.

Remember to pay attention and abide by these properties in your implementation, if you need to serialize Bloom filters and de-serialize them later.

So, if a lookup finds out that one of the bits associated with a key `X` is set to `0`, then we know for sure that `X` was never added to the filter: otherwise all the bits to which `X` is hashed would be `1`.

4.8.2 ...But there are false positives